

Juniorprogrammierer.de

Python 2: Typecasting & Operators

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Agenda

- Most important types
- Typecasting I
- Operators I + II + III
- Homework

Most important types

Example	Data Type
<code>x = "Hello World"</code>	<code>str</code>
<code>x = 20</code>	<code>int</code>
<code>x = 20.5</code>	<code>float</code>
<code>x = 1j</code>	<code>complex</code>
<code>x = ["apple", "banana", "cherry"]</code>	<code>list</code>
<code>x = ("apple", "banana", "cherry")</code>	<code>tuple</code>
<code>x = range(6)</code>	<code>range</code>
<code>x = {"name" : "John", "age" : 36}</code>	<code>dict</code>
<code>x = {"apple", "banana", "cherry"}</code>	<code>set</code>
<code>x = frozenset({"apple", "banana", "cherry"})</code>	<code>frozenset</code>
<code>x = True</code>	<code>bool</code>
<code>x = b"Hello"</code>	<code>bytes</code>
<code>x = bytearray(5)</code>	<code>bytearray</code>
<code>x = memoryview(bytes(5))</code>	<code>memoryview</code>
<code>x = None</code>	<code>NoneType</code>

Typecasting I

- Converting one datatype into another
- For instance, taking something small and put it in something big (e.g. char => String) or the other way around
- Python has one typecasting which is straightforward

```
x = int(1)    # x will be 1
y = int(2.8)  # y will be 2
z = int("3")  # z will be 3
```

```
x = float(1)    # x will be 1.0
y = float(2.8)  # y will be 2.8
z = float("3")  # z will be 3.0
w = float("4.2") # w will be 4.2
```

```
x = str("s1") # x will be 's1'
y = str(2)    # y will be '2'
z = str(3.0)  # z will be '3.0'
```

Operators I

- Arithmetik Operators
 - Plus
 - Minus
 - Multiplator
 - Division
 - Modulus => left over after division
 - Inkrement => adding by 1
 - Dekrement => reducing by 1

```
#-----  
#Operators  
#-----
```

```
u = 10  
w = 3
```

```
r = u + w
```

```
r = u - w
```

```
r = u * w
```

```
r = u / 3
```

```
r = u % w
```

```
#Inkrement
```

```
r += u
```

```
#Dekrement
```

```
r -= w
```

Operators II

- Comparison Operators
 - Equals
 - Not Equal
 - Bigger
 - Smaller
 - Bigger Equal
 - Smaller Equal

```
num = 10  
  
print (num == 10)  
print (num != 10)  
print (num > 10)  
print (num < 10)  
print (num >= 10)  
print (num <= 10)
```

Operators III

- Logical Operators
 - And
 - Or
 - Not

```
print (x < 5 and x < 4)
print (x < 5 or x < 4)
print( not(x < 5 and x < 4))
```

Homework

- Create an arcade.py.
 - You are in an arcade. Int maxScore = 500. You achieved int personalScore = 444. What is your double percentage of maxScore. The formular is $\text{personalScore}/\text{maxScore} * 100$.
- Create an operators_testing.py
 - Look at slides operators I, II, III
 - Play around by writing them down and look at the result